CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME			
CLASS	2T	**INDEX NUMBER**	

BIOLOGY

9648/03

Paper 3 Applications Paper

WEDNESDAY 14 SEPTEMBER 2011
2 Hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all questions.

For Examiner's Use	
1 [13]	
2 [13]	
3 [14]	
Planning	
4 [12]	
Free-response	
5 [20]	
TOTAL	72

This document consists of **14** printed pages.

[Turn over]

Answer **all** questions

1 Fig 1.1 below shows the isolation of a specific human gene and cloning into a bacteriophage.

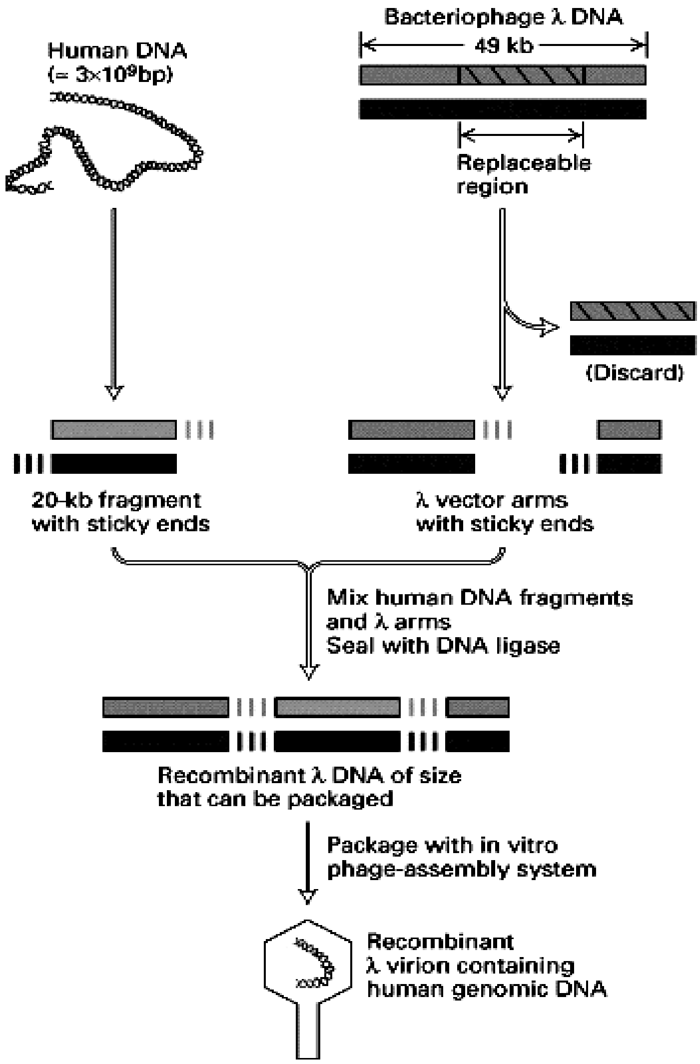

Fig. 1.1

(a) Identify the type of DNA library derived using this procedure.

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………[1]

(b) Name another vector that can be used for the making of DNA libraries. Give two reasons why this vector might be favoured over the bacteriophage vector used here.

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(c) Describe two uses of a cDNA library.

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(d) Name two methods adopted for the screening of a DNA library to identify of clones of interest.

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2: ………………………………………………………………………………………………………

……[2]

Fig. 1.2 shows a plasmid which was isolated from bacteria and engineered to carry 2 selectable markers: *amp*^{r} (confers cell resistance to antibiotic ampicillin) and *lacZ* (encodes enzyme ß–galactosidase). This engineered plasmid is later used in the synthesis of human insulin.

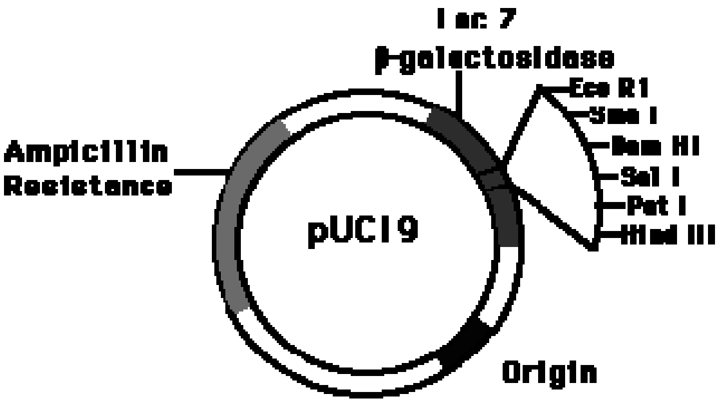

Fig. 1.2

(e) Describe how ampicillin resistance is used in the genetic engineering of human insulin.

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……[2]

Human DNA is digested with the restriction enzyme Bam H1. pUC19 is also digested with the same restriction enzyme at the single restriction site within the *lacZ* gene. The restriction fragments and the cut plasmids are mixed together, allowing base pairing between their complementary sticky ends. DNA ligase is added to seal the nicks. Two types of plasmids will result from this step, namely the re-annealed non-recombinant plasmid and recombinant plasmid containing foreign DNA fragments.

(f) With reference to Fig. 1.2, explain how transformed cells containing recombinant plasmids are differentiated from transformed cells containing re-annealed non-recombinant plasmids.

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[Total: 13]

2 **(a)** Cystic fibrosis (CF) is a genetic disease caused by a mutation in the gene coding for the cystic fibrosis transmembrane regulator (CFTR). The most common mutation is a deletion of three base pairs from the CFTR gene. This mutant allele is carried by 1 in 29 Caucasians in the United States.

The sweat test is the standard diagnostic test for CF. Sweat formation relies on a balance between secretion and reabsorption of sodium chloride by sweat gland cells. A high concentration of sodium chloride in sweat is indicative of CF. Most CF sufferers are diagnosed by the age of 2 years. Some, however, are diagnosed only after 18 years or after.

Fig. 2.1 shows the CFTR protein embedded in the plasma membrane of an epithelial cell. The orientation of the protein differs in different cell types such as sweat gland cells.

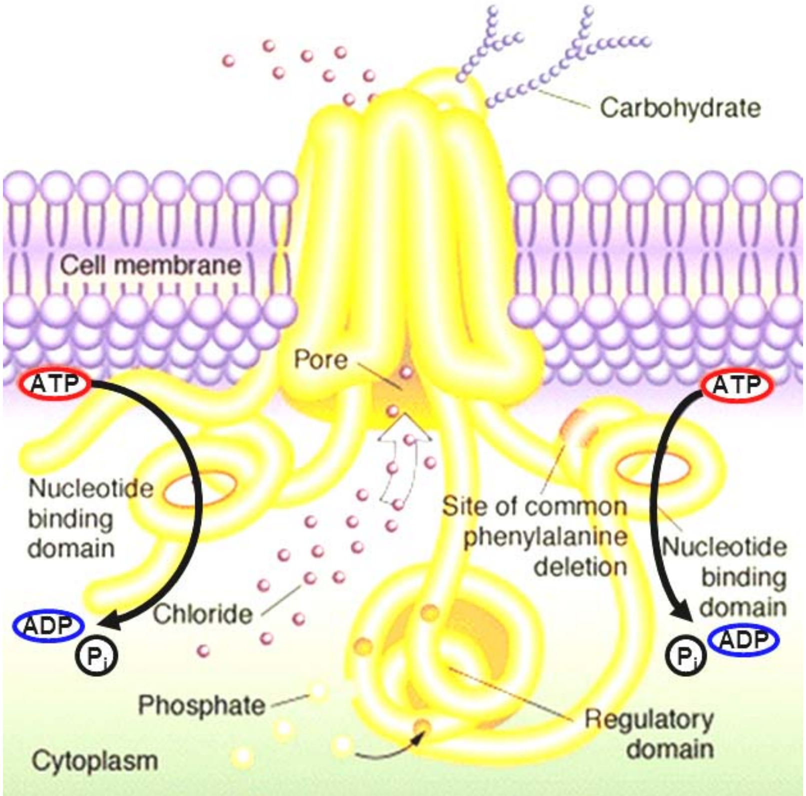

Fig. 2.1

(i) With reference to Fig. 2.1, describe the function of the CFTR protein.

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(ii) Explain how CF sufferers produce sweat with a high concentration of sodium chloride.

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(iii) Suggest why some cases of CF are diagnosed later in life than most.

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(b) CF is currently incurable. Management of the disease is traditionally directed at the symptoms which include poor lung, gastrointestinal and endocrine function. CF sufferers may require lung transplantation as lung function declines. In other diseases, transplantation of a single lung is sufficient even if both lungs fail. CF sufferers must have both lungs replaced.

Gene therapy has provided advances in the treatment of CF. Research has been carried out on the use of viral and non-viral gene delivery vectors.

(i) Suggest why CF sufferers must have both lungs replaced.

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(ii) Describe how liposomes can be used to deliver therapeutic genes and alleles into cells.

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(iii) State one advantage and one disadvantage of an adenoviral vector compared to a liposome vector.

Advantage:

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Disadvantage:

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(c) The film "I Am Legend" (Copyright © 2007 Warner Bros. Pictures) depicts the use of a genetically modified measles-virus as a gene-therapy based cure for cancer.

With reference to the molecular biology of cancer, discuss the feasibility of gene-therapy in treating cancer.

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[Total: 13]

3 **(a)** Plant growth regulators (PGR) such as auxin, cytokinin and ethylene are naturally produced in plants and are important in determination of the developmental pathway of plant cells.

Cheaper synthetic analogues are frequently used in plant tissue culture instead of the natural PGR.

Table 3.1 summarizes the general characteristics and roles of some natural and synthetic PGR.

<table>
<tr><th>Type of PGR</th><th>Examples</th><th>Role</th></tr>
<tr><td rowspan="2">Auxins</td><td>IAA(natural)
- unstable to heat & light</td><td rowspan="2">Promote cell division and cell growth
Root initiation (when auxin:cytokinin is high)</td></tr>
<tr><td>2,4-D(synthetic)
– stable to heat & light</td></tr>
<tr><td rowspan="2">Cytokinins</td><td>2iP(natural)
– unstable to heat & light</td><td rowspan="2">Promotes cell division
Shoot formation (when auxin:cytokinin is low)</td></tr>
<tr><td>Kinetin(synthetic)
- stable to heat & light</td></tr>
<tr><td>Abscisic acid (ABA)</td><td>-</td><td>Inhibits cell division
Maturation of somatic embryos
Abscission (i.e. shedding) of plant leaves
Seed dormancy
Induces stomatal closure to reduce water loss by transpiration</td></tr>
<tr><td>Ethylene</td><td>-</td><td>Abscission (i.e. shedding) of leaves and flowers
Seed and bud dormancy by growth inhibition
Fruit ripening</td></tr>
</table>

Table 3.1

Prunus lannesiana is an early-flowering cherry (only in spring) in Japan Izu peninsula. It is commonly known as Sakura or Japanese Cherry. Researchers are keen to propagate *P. lannesiana* from sterilized explants by micropropagation due to the advantages that the technique offers. Hence, a study was made to analyse the concentration of PGR present in the *P. lannesiana* in the four seasons and the results were summarized in Table 3.2.

Season / Conc. of PGR / arbitrary unit	Spring (Mar-May)	Summer (June – Aug)	Autumn (Sep-Nov)	Winter (Dec – Feb)
IAA (auxin)	15	20	25	30
2iP (cytokinin)	10	15	10	5
ABA	5	5	12	15
Ethylene	12 – 20	18	16	14

Table 3.2

Table 3.3 illustrates the physiological development of *P. lannesiana* in the various seasons.

Season	Spring	Summer	Autumn	Winter
Shoot development	Very Active	Active	Minimal	Nil
Root development	Minimal	Moderate	Active	Very Active
Leaves development	Very Active	Active	Senescence (leaves turning autumn yellow) and abscission	Nil
Flowers development	Short full bloom in early spring, followed by senescence	Nil	Nil	Buds develop but dormant in late winter
Fruit and seed development	Fruit and seed development in late spring	Seed maturation	Seed dormant	Seed dormant

Table 3.3

(i) State two limitations of micropropagation.

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2: ……………………………………………………………………………………………...

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(ii) Suggest why synthetic PGR (e.g. 2,4-D and Kinetin) are used instead of natural PGR (e.g. IAA and 2iP) in plant tissue culture.

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(iii) Using the information provided in Table 3.2 and Table 3.3, explain the effect of the change in the auxin:cytokinin ratio from Spring to Winter and vice versa.

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(b) The Salmon Genome Project (SGP) is developed to increase knowledge of the biology of Atlantic salmon and aid agricultural breeding of the fish.

(i) Describe how the SGP serves to increase knowledge of the biology of Atlantic salmon.

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………[2]

In order to produce transgenic salmon expressing salmon growth hormone, sGH, the coding sequence of sGH gene is isolated from a library and cloned, using *Sac* I, into the plasmid expression vector, pBluescript II SK. The plasmid map is shown in **Fig. 3.1**.

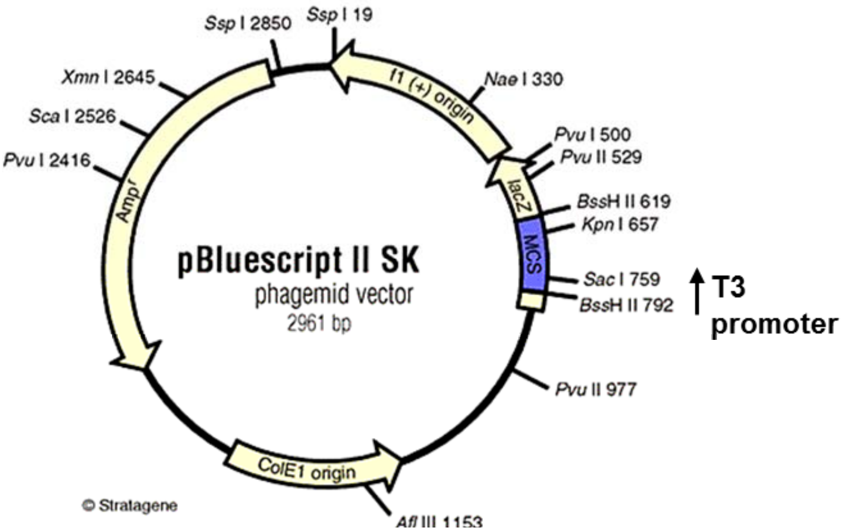

Fig. 3.1

(ii) With reference to Fig. 3.1, describe the features of the multiple cloning site (MCS).

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Subsequent to insertion of sGH gene into pBlueScript II SK, transformation into *E. coli* cells was carried out and some colonies were obtained. The plasmid DNA was extracted, digested with *Sac* I and the restriction fragments were separated in gel electrophoresis.

(iii) It was found that the recombinant plasmid with sGH gene inserted did not yield the expected polypeptide. With reference to Fig. 3.1, state a reason and provide an explanation for this.

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(iv) Suggest a solution for the problem in **(iii)**.

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[Total: 14]

Planning question

4 Fig. 4.1 shows a colorimeter that can be used to measure the absorbance of particular wavelengths of light by a solution. Different solutes absorb different wavelengths of light. The higher the concentration of a particular solute, the greater corresponding wavelength of light is absorbed.

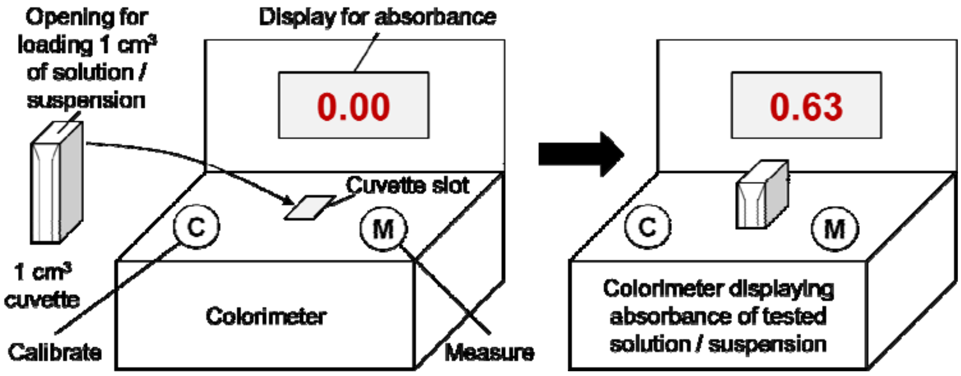

Fig. 4.1

Acidified $KMnO_4$ absorbs light with a wavelength of 525 nm. It can be reduced to Mn^{2+} as follows:

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

Design an experiment to test the hypothesis that:

The rate of photosynthesis is dependent on the intensity of light.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which *you* must use:

- colorimeter with 1 cm^3 cuvettes
- fresh spinach leaves
- sharp knife
- mortar and pestle
- leaf extract buffer solution
- ice
- a light-proof container with a removable lid
- light-proof aluminium foil
- acidified potassium permanganate solution (freshly made)
- 40 W lamp (chosen so it does not generate heat)
- ruler
- fine sharp sand
- a room which can be made dark using curtains
- a variety of different sized beakers, measuring cylinders, syringes and pipettes for measuring volumes

A leaf extract can be made by mixing finely ground leaf with cold buffer solution. This leaf extract should be kept cold and in the dark except when taking samples.

[Total: 12]

Your plan should have a clear and helpful structure to include:
- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken.

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Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Describe how gel electrophoresis and nucleic acid hybridisation are carried out for DNA analysis. [8]

(b) Explain the advantages and limitations of the polymerase chain reaction as a tool in molecular biology. [6]

(c) Discuss the social and ethical implications of the Human Genome Project. [6]

[Total: 20]

END OF PAPER